

Psychomotor Stimulants

Cocaine, Amphetamine, Methylphenidate
Chapter 12

3 Key points

- 1 The drug actions of cocaine & m/Amph
- 2 The behavioral effects of cocaine & m/Amph
- 3 Neuroadaptations following chronic use
 - Craving
 - Tolerance & Sensitization
 - Hypofunction of PFC

Cocaine is derived from *Erythroxylon coca*



Franz Eugen Köhler, Köhler's Medizinal-Pflanzen [Public domain]
https://upload.wikimedia.org/wikipedia/commons/4/42/Erythroxylum_coca_-_Franz_Eugen_Köhler%26Zeno%26Giovanni_Medizinal-Pflanzen-204.jpg

cocaine-HCl

- oral, intranasal, IV
- vulnerable to heat

free base

- dissolve in H₂O, NaHCO₃, heat & dry = "crack"

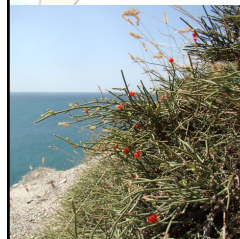
lipophilic

- easily crosses BBB

metabolites

- benzoylecgonine
- coc+EtOH→cocaethylene

Ephedrine is derived from *Ephedra vulgaris*



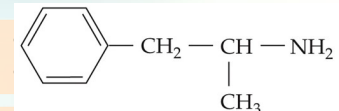
Yury Danilevsky [CC-BY 4.0]
<https://www.pbrf.org/occurrence/1993739132>

ephedrine
pseudoephedrine

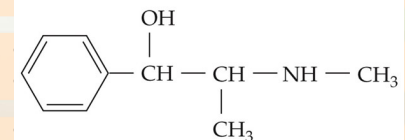
amph

meth

metabolism



Amphetamine



Ephedrine

PSYCHOPHARMACOLOGY 4e, Figure 12.13
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The primary drug actions of cocaine are

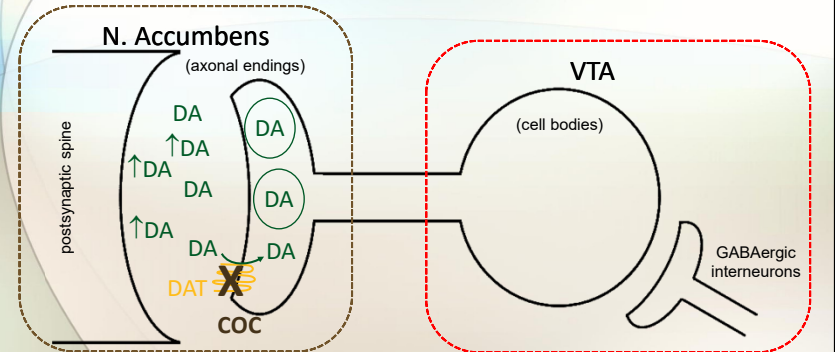
inhibition of neurotransmitter transporters

- low affinity for DAT
- higher affinity for NET & SERT

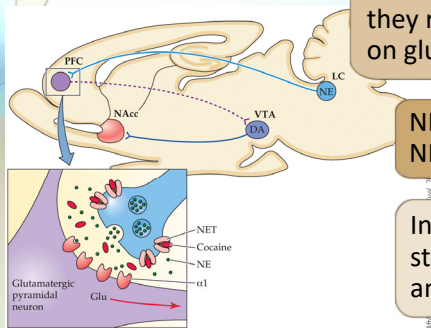
increased DA release independent of DAT

- inhibition of NET in PFC drives excitation of VTA

The primary drug action of cocaine is inhibition of DAT (& NET, & SERT)



Cocaine increases DA release independent of actions on DAT



LC fibers project to the PFC, where they release NE onto excitatory $\alpha 1$ -AR on glutamatergic pyramidal neurons.

NET inhibition by COC increases
NE action at $\alpha 1$ -AR in PFC.

Increased excitation in PFC stimulates DA neurons in the VTA and elevates release of DA in NAcc

The primary drug actions of cocaine are

inhibition of neurotransmitter transporters

- low affinity for DAT
- higher affinity for NET & SERT

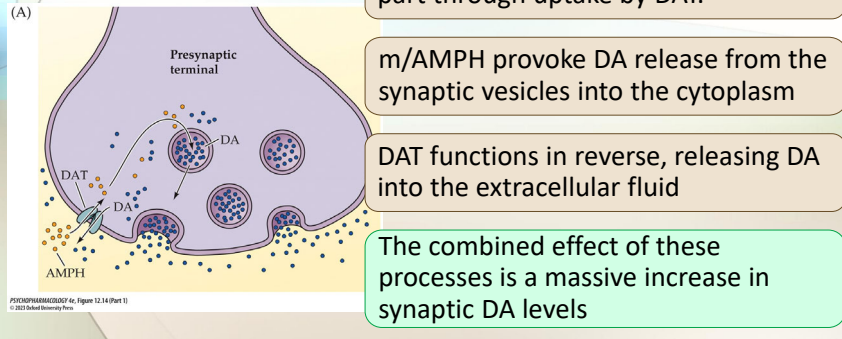
increased DA release independent of DAT

- inhibition of NET in PFC drives excitation of VTA

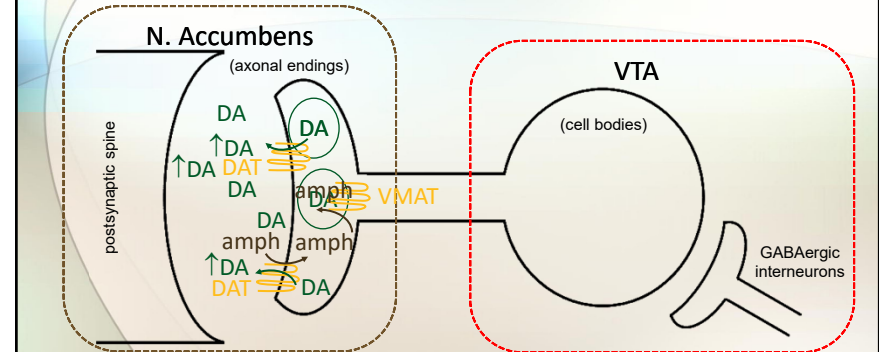
inhibition of VGNaC (local anesthetic)

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AMPH & mAMPH disrupt the monoamine transport cycle



AMPH is a substrate for both DAT & VMAT



The primary drug actions of AMPH/mAMPH are

inhibition of neurotransmitter transporters

- DAT, NET & SERT (similar to COC)

increased DA release by reversal of DAT

- AMPH & mAMPH are substrates for VMAT & DAT
- m/AMPH are taken into vesicles, disrupt H^+ gradient, reduce transport of DA into vesicles by VMAT; $[DA]_{cyto}$ increases
- DAT runs in reverse, releasing DA into synapse

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The behavioral effects of COC include

Euphoric “high”

- exhilaration
- more energy
- less fatigue
- reinforcing
- self-administered

Behavioral activation

- repetitive, stereotyped behaviors
- locomotion, rearing, sniffing

Sympathetic arousal

- tachycardia
- vasoconstriction
- hypertension
- hyperthermia

AMPH also produces other effects

Other behavioral effects

- increased alertness, decreased sleep & improved performance
- sustained physical effort without need for sleep
 - truck drivers, military, athletes

Therapeutic uses

- Weight loss
- ADHD
- Narcolepsy

The drug effects of psychostimulants depend on mesolimbic DA systems

Experimental manipulation	Brain area	Behavioral effect
Psychostimulant microinjection	Nucleus accumbens	Increased locomotor behavior
Psychostimulant microinjection	Dorsal striatum	Increased stereotyped behaviors
6-Hydroxydopamine (6-OHDA) lesion	Nucleus accumbens	Decreased locomotor response following systemic administration of a low-dose psychostimulant
6-OHDA lesion	Dorsal striatum	Decreased stereotyped behaviors following systemic administration of a high-dose psychostimulant
6-OHDA lesion	Nucleus accumbens	Decreased reinforcing effectiveness of systemically administered psychostimulants
Psychostimulant microinjection	Nucleus accumbens*	Reinforcing to the animal

COC-insensitive DAT-ki mice do not self-administer COC

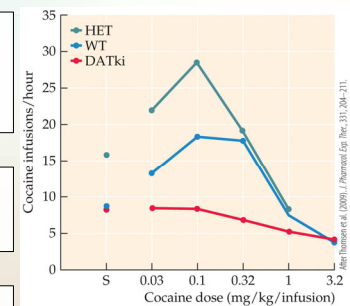
DAT “knock-in”

- transports DA
- but not blocked by COC

WT and HET mice

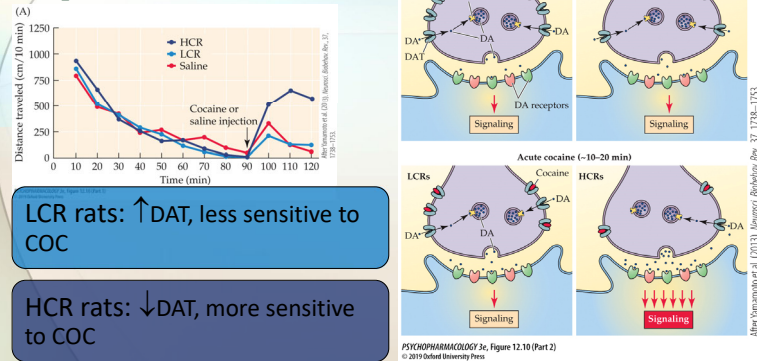
- show dose-dependent SA of COC

DAT-ki mice do not



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Variations in DAT expression underlie behavioral responses to COC



More DA, so what? Which receptor(s) are most important in the behavioral effects of COC?

5 different DA receptors

- D1 & D5
- D2, D3 & D4

D1 KO

- required for COC-induced behavioral activation
- do not self-administer COC

D2 or D3 KO

- not required for behavioral effects of COC
- persist in COC self-administration

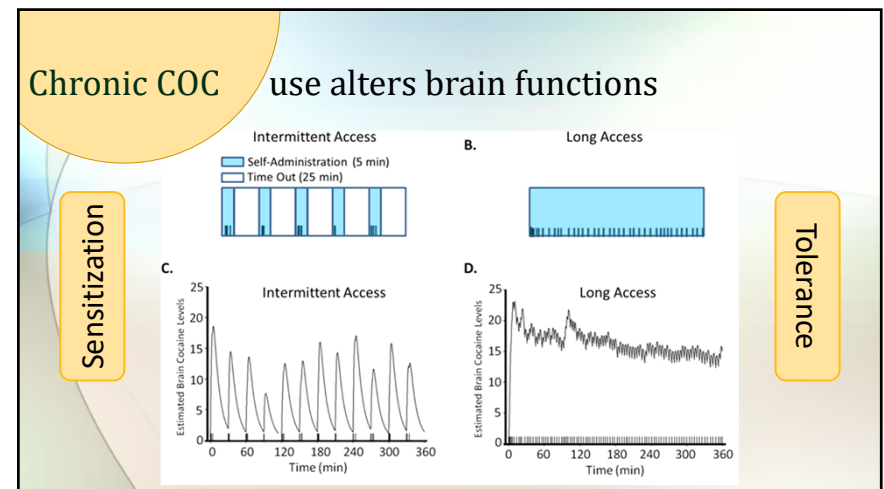
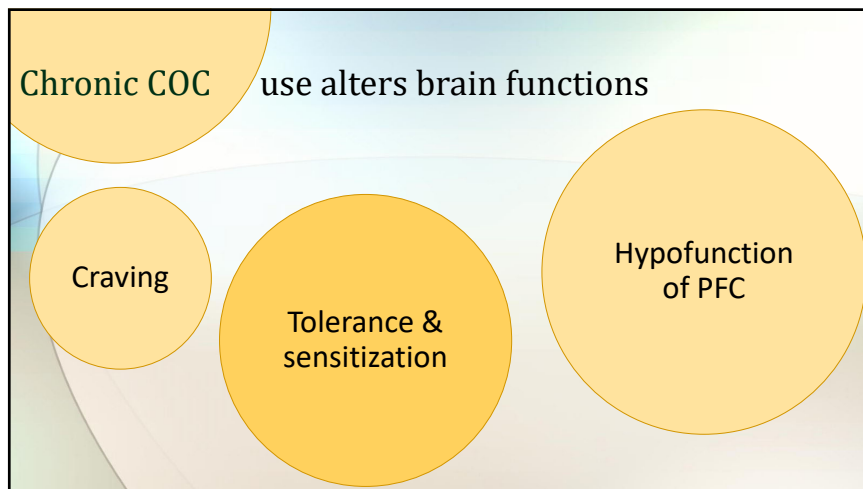
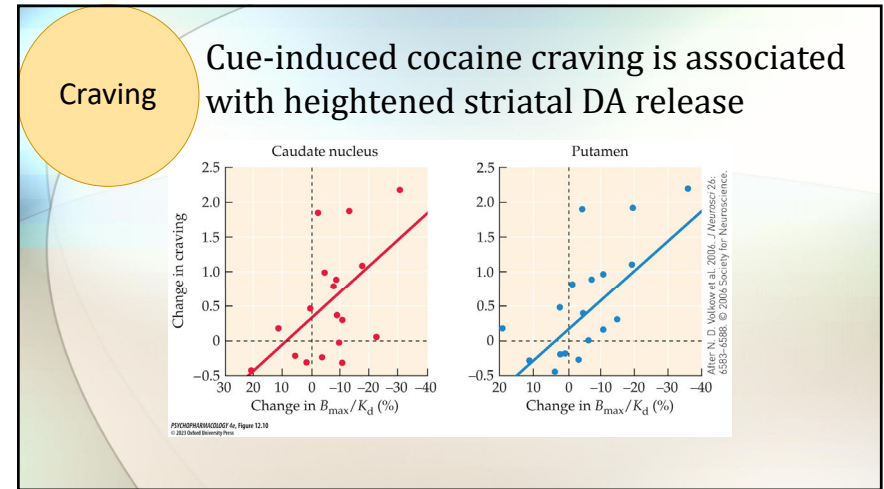
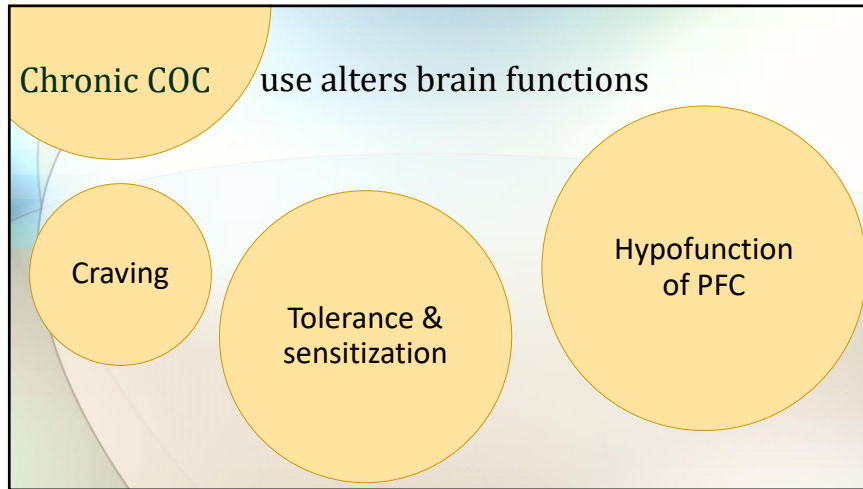
Interim summary

- Cocaine, amphetamine and methamphetamine are psychomotor stimulants with strong abuse potential
- Stimulants are rewarding (reinforcing), and are self-administered when administered systemically or infused directly into N. Acc.
- Drug effects are largely the result of excess synaptic DA (and DA release) due to drug actions on DAT (as well as NET & VMAT). D1 receptor activation is required.

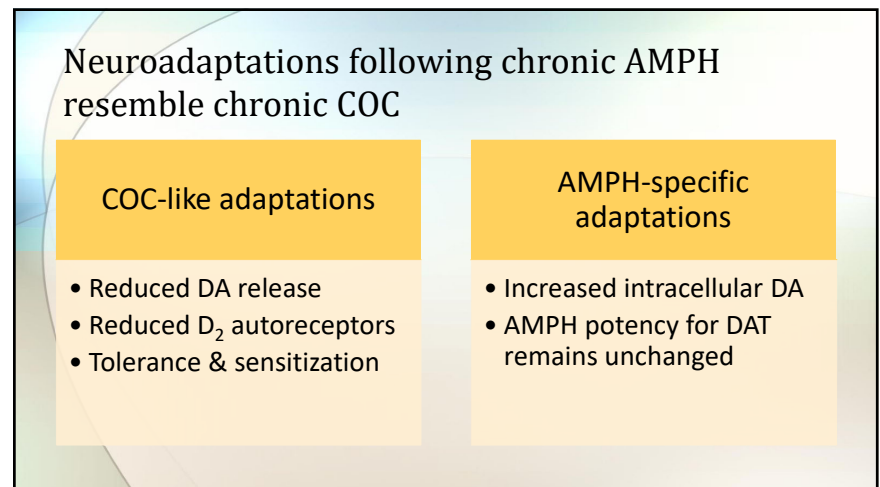
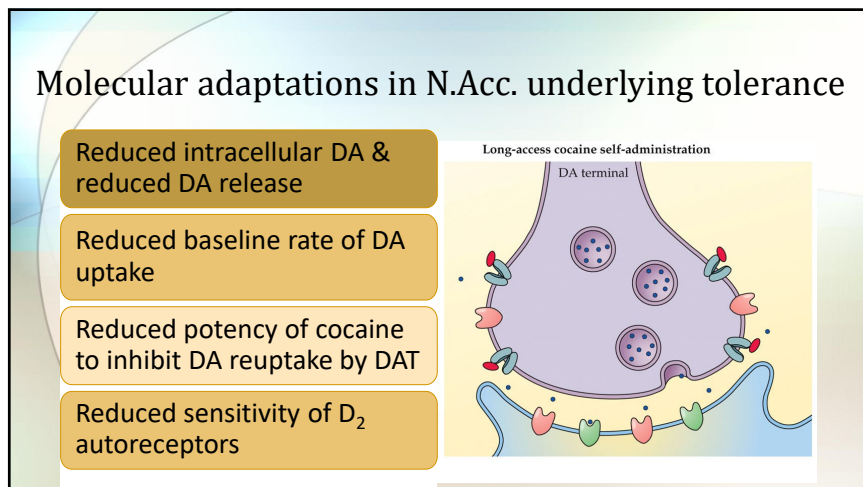
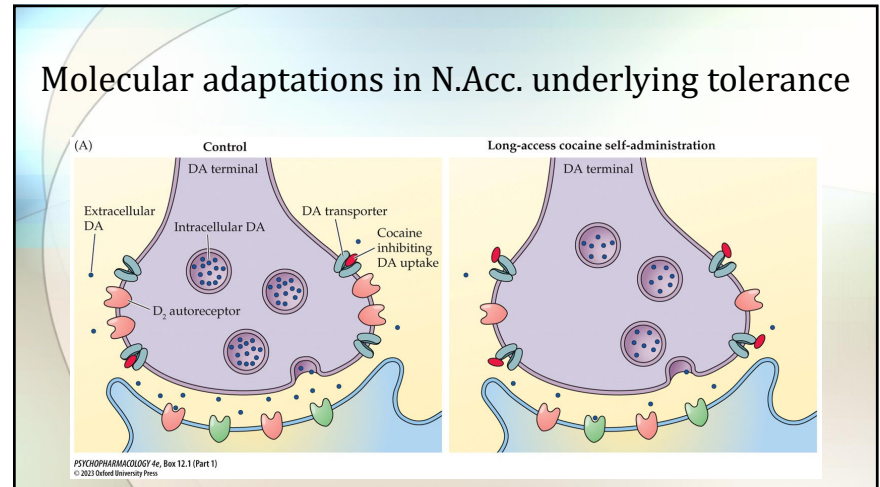
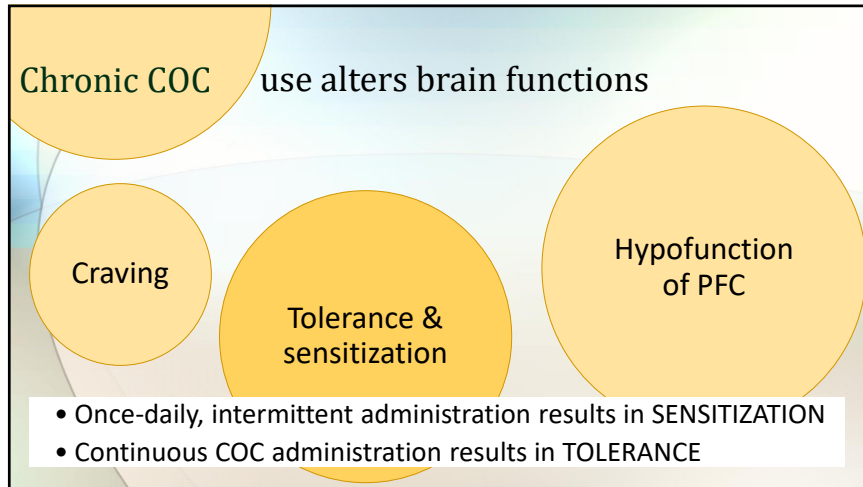
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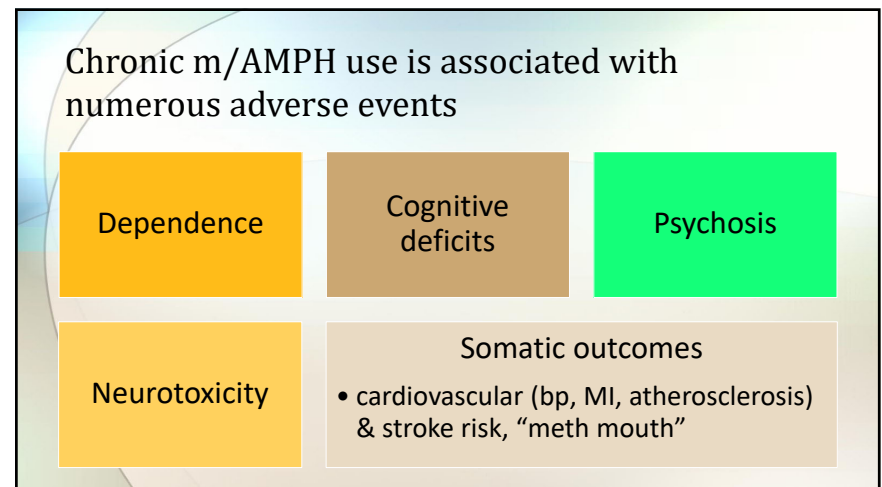
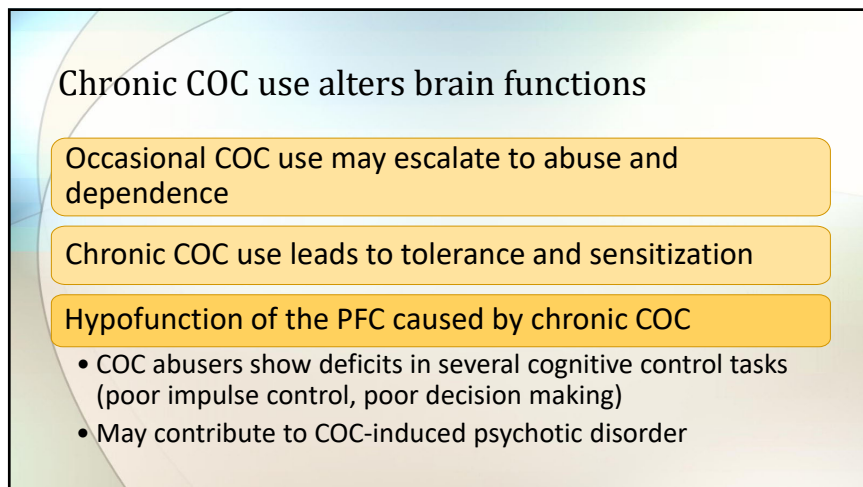
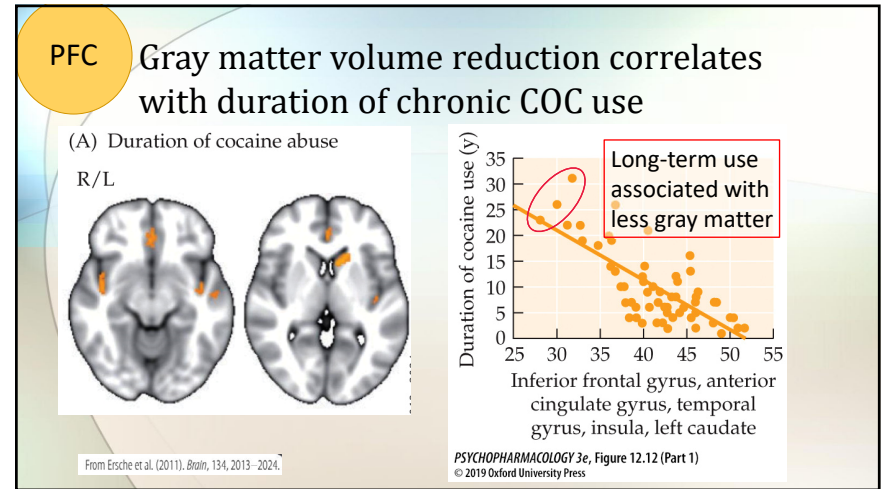
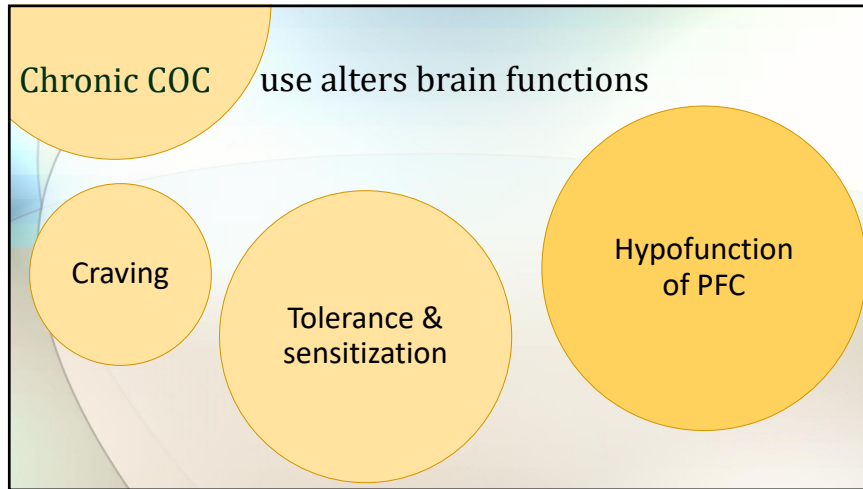
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Interim summary

Mesolimbic DA

- Changes to mesolimbic DA system contribute to enhanced craving, tolerance & sensitization

PFC deficits

- Chronic stimulant use is associated with functional deficits in the PFC
- Loss of control over subcortical “habit” systems; Stimulant-induced psychosis

Psychostimulants (paradoxically) improve the symptoms of attention deficit / hyperactivity disorder (ADHD)

ADHD medications

- amphetamine
- methylphenidate
- α_2 AR agonists
- NRI

